

SHUVALAXMI DASS, PHD

Website

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SUMMARY

- Dr. Dass is a CEH-certified interdisciplinary researcher with expertise in cybersecurity, evolutionary algorithms, machine learning, and reinforcement learning. Dr. Dass is currently studying the impact of integrating different evolutionary algorithms with Reinforcement Learning in developing frameworks used for securing misconfigured software applications from attacks.
- Dr. Dass currently serves as an assistant professor University of Louisiana at Lafayette (UL). She also holds the position of IEEE chair of Computer Society, Lafayette chapter and is an active member of Women in Cybersecurity (WiCys). She recently launched the Ethical Hacking Club in UL serving as the faculty advisor. Her research interests are Software Security, ML, Vulnerability Analysis, Bio-inspired Algorithms, privacy.

EDUCATION

- PhD Computer Science, Texas Tech University, TX, June 2021
- MS, Software Engineering, Gautam Buddha University, India, May 2017
- BS, Computer Science, Gautam Buddha University, India, Dec 2015

PUBLICATIONS

- Mohammad Farhad, Sabbir Rahman, and **Shuvalaxmi Dass**. "HYDRA: A hybrid heuristic-guided deep representation architecture for predicting latent zero-day vulnerabilities in patched functions." *Accepted in IEEE Automated Software Testing (AST) 2026*.
- Valentin Leroy, **Shuvalaxmi Dass**, Sharif Ullah. "Memory-Based Malware Detection under Limited Data Conditions: A Comparative Evaluation of TabPFN and Ensemble Models." *Accepted in 2026 Workshop on Computing, Networking and Comms. (CNC)*.
- Niloofar Heidarikohol, **Shuvalaxmi Dass** "Advancing Moving Target Strategy with Bio-Inspired Reinforcement Learning to Secure Misconfigured Software Applications". *Journal of Artificial Intelligence & Cloud Computing*. 2025, SRC/JAICC-523.
- Bou Abdo, J., **Dass, S.**, Qolomany, B., Hossain, L. (2024). *Modeling the Dynamics of Bitcoin Overlay Network*. In: Cherifi, H., Rocha, L.M., Cherifi, C., Donduran, M. (eds) *Complex Networks & Their Applications XII*. COMPLEX NETWORKS 2023. *Studies in Computational Intelligence*, vol 1143. Springer, Cham.
- J. Schnell, **S. Dass**, J. B. Abdo and M. Chatterjee, "Cultivating Inclusivity: A Journey Towards Democratizing Cybersecurity Education," *2023 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*, Auckland, New Zealand, 2023, pp. 1-4, doi: 10.1109/TALE56641.2023.10398317.
- Stephanie Abanilla, Moitrayee Chatterjee, **S. Dass**, "Towards Privacy-Preserving Financial Fraud Detection", *2023 International Conference on Machine Learning and Applications (ICMLA)*, Jacksonville, FL, USA, 2023, pp. 1723-1726
- Serge Sandakly, Charbel Salem, Khalil Challita, **Shuvalaxmi Dass**, et al. XorshiftH128+: A hybrid random number generator for lightweight IoT, *2023 IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings)*.
- **Shuvalaxmi Dass**, Akbar Siami Namin. Reinforcement Learning for Generating Secure Configurations. *Electronics* 2021, 10, 2392. <https://doi.org/10.3390/electronics10192392>
- **Shuvalaxmi Dass**, Akbar Siami Namin. Vulnerability coverage for adequacy security testing, pp. 540-543, *SAC '20: The 35th ACM/SIGAPP Symposium On Applied Computing 2020*
- **Shuvalaxmi Dass** and Akbar Siami Namin. "Evolutionary Algorithms for Vulnerability Coverage." *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)*, Madrid, Spain, 2020, pp. 1795-1801
- **S. Dass** and X. Xue and A. Siami Namin. "Ensemble Random Forests Classifier for Detecting Coincidentally Correct Test Cases," *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)*, Madrid, Spain, 2020, pp. 1326-1331, doi: 10.1109/COMPSAC48688.2020.00-72.
- **Shuvalaxmi Dass** and Akbar Siami Namin, "A Sensitivity Analysis of Evolutionary Algorithms in Generating," *2020 IEEE International Conference on Big Data (Big Data)*, Atlanta, GA, USA, 2020, pp. 2065-2072, doi: 10.1109/BigData50022.2020.9378307.
- **Shuvalaxmi Dass**, Prerit Datta and Akbar Siami Namin, "A Hidden Markov Model for Attack Prediction", *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, Madrid, Spain, 2021, pp. 1695-1702, doi: 10.1109/COMP-SAC51774.2021.00253.

GRANTS

1. NSF SaTC RES [PI]: MTD-CH: A Moving Target Defense - OS Configuration Hardening Framework for Real-Time Mitigation of Configuration Drift and Zero-Day Attacks. [Under Review]
2. NSF CyberAI Innovation [PI]: SeCURE-CPS - AI and Security education through CUREs for Cyber-Physical Systems [Under Review]
3. Amazon Research Award RFP [PI]: AEGIS: An Agentic AI Framework for Automated Zero-Day Vulnerability Prediction and Remediation in Patched Code. [Under Review]
4. NSF EPSCoR and LA-NEXT Graduate Student Supplements (GRASS) [PI]: ZeroAware : AI-Integrated Zero-Day Vulnerability Prediction in CI/CD pipeline for Critical infrastructure. [AWARDED - \$10K]
5. ReU UNK internal Grant [PI]: Visual Modelling Techniques for Cybersecurity Concepts. [Funded - 5000\$]

PROFESSIONAL EXPERIENCE

University of Louisiana at Lafayette, Louisiana

Assistant Professor

Aug 2023 to Present

1. Research and Teaching (INFX 422- Ethical Hacking, INFX 455- Cyber Physical System security, INFX 301-Computation Thinking)

University of Nebraska at Kearney, Nebraska

Assistant Professor

Aug 2021 to July 2023

1. Designed and taught 6 new UG CS courses: CYBR 100, CYBR 150, CYBR 180, CYBR 158, CYBR 401, CYBR 140
2. Designed new course for CYBR 800 which is a grad level course of CYBR 100 for teachers ed in K12 schools.

Texas Tech University, Texas

PhD student

Aug 2017 to Aug 2020

1. Taught supervised and unsupervised ML algorithms, statistical methods like Hidden Markov models to undergraduate research group students of 10
2. Taught introductory lecture on Deep Learning to a class of 20 graduate students as a guest lecture.[Class Recording]
3. Contributed to an NSF funded educational website: Wikiset in developing lab tutorials on Android Forensics.

MENTORING EXPERIENCE

- **French exchange summer students, CESI:** Advisor - 2024
 - Anya: Biometric security.
 - Gabriel: Social Engineering security.
 - Benoit: Malware security.
- **Graduate Student, UL at Lafayette:**
 - Sabbir[2025-]: PhD student
 - Farhad[2025-]: PhD student
 - Mohoshin[2024-]: Master's Thesis, Topic: LLM in healthcare
 - Niloofar[2023]: Independent study, Topic: Designing particle swarm optimization technique for optimizing Reinforcement Learning Based Moving target Defense.
- **Undergraduate Students, UNK:** Mentored following students in security-related research: 2021-2022
 - Jordan Schnell: Co-Designed and co-developed Visualization Tool dashboard for CIA security triad.
 - Ramakrishna Gurung: Mentored him in designing an evolutionary-inspired game-based Reinforcement Learning algorithm for configuration security.
- **Undergraduate Students, TTU:** Mentored several CSE students to provide research and team-project experience:
 - 2020: Mentored 4 undergraduates on their project on Machine Learning.
 - 2019: Mentored 6 undergraduates on their project on Hidden Markov Models.

INVITED TALKS

- Invited speaker at 2025 International Conference on AI (ICAI)
- Invited speaker at 2024 International Conference on Artificial Intelligence and Machine Learning. (AIM)
- Virtual speaker to deliver a talk on my research focus titled "Achieving Configuration security through Moving Target Defense" at 2023 IEEE ComSoc Lebanon.
- Hosted a UG seminar talk on "Application of AI in cybersecurity" at College of Engineering, Texas A&M University, College Station 2023
- Speaker to talk on "Cybersecurity awareness" for Kearney Women of Today club in Nebraska 2022 (outreach)

SERVICE

- **2025:** Contributed to the development of a new UG course on INFX - 100 - Essentials and Applications of Artificial Intelligence and Machine Learning
- **2023:**
 - Volunteered senior faculty in organizing Science Fair Day at UL for 1000 high school students from Lafayette and nearby.
- **2022:**
 - Invited as Speaker on Cybersecurity Awareness day for Kearney Women of Today.
 - Involved in the on-boarding process and orientation of incoming students in cyber systems department.
 - Student Advising: Help students keep track of their progress towards degree completion through timely course enrollment.
 - Chairing Student Affairs committee. Duties include:
 - * Led and contributed to the creation of a student survey using Qualtrics for the Dean to use in order to gain insight into students' struggles with Academic, Advising, and Support.
 - * Assigning Scholarship disbursement funds to students based on multiple factors.
- **2021:**
 - Participated in meetings related to department's program level changes.
 - Introduced and Developed a new course CYBR 800 for teaching programming logic for K12 teachers ed
 - Incorporated Python into CYBR 150, which was initially offered in Java, to ensure student retention by making the transition from CYBR 101: Python Basics smooth.
 - Committee member of Student Affairs committee. Duties include:
 - * Interviewed UNK students in the Living Learning Community (LLC) to understand their living conditions and unmet needs.
 - * Contributed to the amendments/actions that need to be made/taken to improve the LLC for the Dean.
 - * Assigning Scholarship disbursement funds to students based on multiple factors.

REVIEWERS

- **2026** TPC IEEE-IMCET'26 (IEEE Lebanon Section)

- **2024** - External Reviewer for IEEE Transactions on Industrial Informatics.
- **2022** - Grant Reviewer of Natural Sciences and Engineering Research Council of Canada (NSERC)
- **2020** - External Reviewer for Hindwai 2020.
- **2019**- External Reviewer for IEEE BigComp 2019.

HONORS AND AWARDS

- **2025** - Nominated to participate in CCC Symposium with travel grant
- **2024** - Proposal Summary Draft selected and invited for a competitive Aspiring PI NSF SaTC workshop scholarship out of 250 applicants.
- **2024** - Recipient of Faculty Scholarship Award for 2024 Women in Cybersecurity Conference (WiCyS).
- **2024** - Selected and invited for 2024 CRA Career Mentoring Workshops (CMW).
- **2022, 2019** - Recipient of CRA-W Grad Cohort Workshop for Women Scholarship travel grant.
- **2017** - Graduate Fellowship awarded by AT&T for 4 years.
- Recipient of Summer Research Fellowship (SRF) Stipend for Summer Internship Program at Indian Institute Technology (IIT) Delhi 2015 & 2016.
- Secured overall 38th rank in Gautam Buddha University(GBU) entrance examination.
- Secured overall 5th rank upto 4 years in GBU(Bachelors)

CERTIFICATIONS

EC-Council: Certified Ethical Hacker (CEH) - Issued Dec 2022

OTHER PROFESSIONAL EXPERIENCE

Truno Retail Technology Solutions,Texas

Data Scientist Intern

May 2019 to Aug 2019

- Worked on forecasting sales on department level for recently opened Sprouts supermarket all over the US that has limited historical sales.
- Worked on forecasting unusual sales during festive and national holidays.
- Developed multiple clustering models: Agglomerative, K-means clustering to cluster new stores with old stores to improve the forecasting accuracy of sales.

Department of Management Studies, IIT Delhi

Summer Intern

May 2016 to July 2016

- Assisted in creating and designing layout for a web based online system
- Developed using SQL Server RDBMS and .Net Programming Language.
- The website served as a job finding platform where IIT students can browse through multiple companies and upload CVs.

Department of Computer Science and Engineering, IIT Delhi

Summer Intern

May 2015 to July 2015

- Assessed the page load time of the web sites with and w/o CDNs using statistical measures including Linear Regression.
- Successfully discovered that websites possessing CDNs are easily and quickly accessed by rural users.